

Jackson “Jack” Houck

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EDUCATION

Boston College - Morrissey College of Arts and Sciences

Bachelor of Arts in Mathematics, Bachelor of Science in Computer Science

Chestnut Hill, MA

Expected May 2028

- **GPA:** 3.75/4.00; **ACT:** 35
- **Honors:** Dean's List (Every Semester)
- **Relevant Coursework:** Machine Learning, Natural Language Processing, Probability, Mathematical Statistics, Linear Algebra, Differential Equations, Multivariable Calculus, Data Structures, Foundations of Blockchain

EXPERIENCE

Natural Language Processing REU Researcher

University of Wisconsin-Milwaukee

Milwaukee, WI

Jun. 2026 – Present

- Extended nanoGPT's PyTorch implementation of GPT-2 using token-aligned linguistic metadata at the embedding layer, reducing validation bits-per-byte by 3.6% against a compute-matched baseline on encyclopedic corpora
- Added a multi-task annotation-prediction head trained with binary cross-entropy alongside the language-modeling task, enabling the model to predict structural metadata at inference without requiring external annotations
- Built an end-to-end pipeline for parsing nested markup, aligning subword annotations, and loading dual-stream memmaps; scaled training to multi-GPU SLURM jobs on Purdue Anvil through an ACCESS-CI allocation
- Presented at the UW-Milwaukee Summer Research Symposium; first author on a manuscript in preparation

Executive Board Member

Heights Capital Corporation

Chestnut Hill, MA

Aug. 2024 – Present

- Served as an Executive Board analyst in the Macro, Quantitative, and Fixed Income Pod, delivering weekly market updates, analyzing economic releases, monetary-policy developments, and major market movements
- Led over 30 candidate interviews and collaborated with peers to design a comprehensive educational curriculum, organize alumni networking chats, and coordinate social events that strengthened overall member engagement

Undergraduate Teaching Assistant

Boston College Department of Computer Science

Chestnut Hill, MA

Aug. 2025 – Jan. 2026

- Instructed beginner students in Computer Science 101 (Python) as a TA by leading weekly discussion sections and collaborative office hours, guiding students in writing their first programs and developing debugging skills
- Graded over 40 assignments weekly and provided actionable feedback to help students strengthen coding style, logical structure, and problem-solving strategies, resulting in measurable improvement in project quality

PROJECTS

HER2 Breast Cancer Image Classification | *Python, PyTorch*

Mar. 2026 – May 2026

- Developed a ConvNeXt-Tiny pipeline for four-class HER2 scoring from immunohistochemistry images, supporting patch-level training, whole-slide-style inference, and spatial model interpretation
- Designed a learned heatmap head with top-*k* pooling and a custom loss combining DAB-stain alignment, sparsity, and border penalties to localize relevant tissue while suppressing shortcut features

Options Volatility Surface Modeling | *Python, scikit-learn, PyTorch*

Feb. 2026 – Apr. 2026

- Engineered a reproducible options-data pipeline that transformed raw SPY and QQQ contracts into model-ready features including log-moneyness, time to expiration, volatility-index levels, and Treasury yields
- Compared random forest, gradient-boosting, multilayer-perceptron, and TabNet regressors for implied-volatility prediction; serialized trained models and scalers for reusable inference
- Developed evaluation and visualization tooling to compute contract-level absolute errors and reconstruct predicted volatility surfaces across strike and maturity dimensions

TECHNICAL SKILLS

Languages & Packages: Proficient: Python, NumPy; Familiar: Java, C++, scikit-learn, pandas, PyTorch

Activities: Eagle Scout, BC Blockchain (Vice President), Intramural Hockey & Soccer

Interests: SciFi Novels, Homelabbing, European History, Warhammer 40k, Boston College Hockey, Formula 1